

BIOLOGY PRACTICAL

Professional Marking Scheme Solutions

Based on the confirmed specimen list supplied

Prepared and compiled by MR OYIBO CEO

Specimen	Identity
A	Cockroach - whole, wet preserved
B	Egg case of cockroach
C	Grasshopper - whole, wet preserved
D	Boiled maize seeds
E	Fresh and viable maize seeds
F	Weevil
G	Powdered milk
H	Piece of bread
J	Cactus plant
K	Waterleaf plant
L	Scapula of a small mammal
M	Humerus of a small mammal
N	Femur of a small mammal

General marking note: Award marks for any correct observable biological point. Do not penalize candidates for using equivalent terms such as maize grain/seed, shoulder blade/scapula, upper arm bone/humerus, or thigh bone/femur.

QUESTION 1: SPECIMENS A AND C

Specimen A = Cockroach; Specimen C = Grasshopper.

1(a)(i) Three observable similarities between the wings of specimens A and C [3 marks]

- Both possess two pairs of wings: forewings and hind wings.
- Both wings are attached to the thoracic region of the body.
- Both have thin membranous hind wings used for flight.
- Both wings show veins/vein-like supports.

1(a)(ii) Three observable differences between the wings of A and C [3 marks]

A - Cockroach	C - Grasshopper
Forewings are leathery, narrow and overlap the abdomen.	Forewings are longer, tougher/tegmina-like and cover the folded hind wings.
Hind wings are broad, thin and folded fan-like beneath the forewings.	Hind wings are larger, more conspicuous and mainly used for strong flight.
Wings generally lie flat over the body.	Wings are held more roof-like over the body when at rest.

1(b) Function of forewing and hind wing of specimen A [2 marks]

- Forewing: protects the delicate hind wing and body; also helps slightly in gliding.
- Hind wing: used mainly for flight.

1(c) One habitat each of specimens A and C [2 marks]

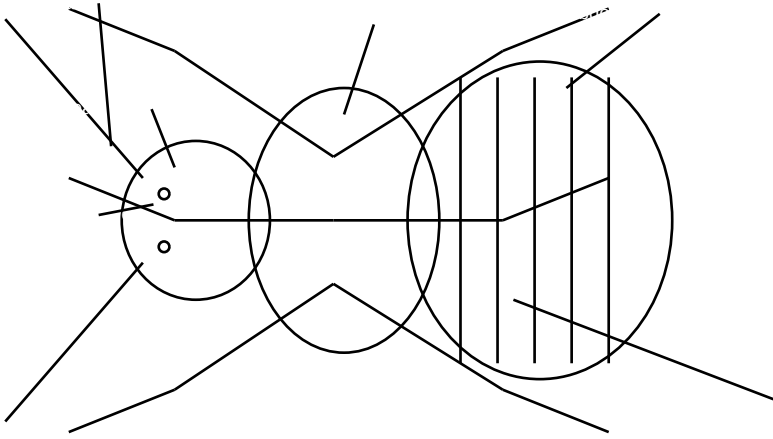
- A - dark, warm and damp places such as kitchens, drains, cupboards, stores and refuse areas.
- C - grassland, crop farm, field, bush or vegetation.

1(d) Three adaptations of specimen C to its habitat [6 marks]

- Powerful hind legs with enlarged femur for jumping among grasses and escaping predators.
- Green or brown body colour provides camouflage among grasses and vegetation.
- Biting and chewing mouthparts enable it to feed on leaves and tender plant parts.
- Wings enable flight from one plant to another and escape from enemies.
- Spiracles and tracheal system allow gaseous exchange on land.
- Claws/adhesive pads help it hold firmly onto plant surfaces.

1(e) Drawing guide: dorsal view of specimen A after removal of wings [9 marks]

- Drawing should be 6 cm - 8 cm long, clear, single-lined and without shading.
- Correct body division must show head, thorax and segmented abdomen.
- Label fully: antenna, compound eye, head, thorax, abdomen, walking legs and cerci where visible.
- Label lines should be straight, not crossing, and labels should be written horizontally.



QUESTION 2: SPECIMENS D, E AND F

D = Boiled maize seeds; E = Fresh viable maize seeds; F = Weevil.

2(a)(i) Relationship between specimens E and F [1 mark]

- Specimen F is a pest/parasite of specimen E; it feeds on and damages maize seeds.

2(a)(ii) Habitat of specimen F [1 mark]

- Stored grains, especially stored maize seeds in barns, silos, sacks or grain stores.

2(b) Observable features of specimen F adapting it to the relationship [3 marks]

- Possesses a hard exoskeleton which protects it inside stored grains.
- Has biting and chewing mouthparts/rostrum for boring into and feeding on maize seeds.
- Small body size enables it to enter holes and spaces between stored grains.
- Has claws/legs for clinging to grains.
- May have wings for dispersal from one store to another.

2(c)(i) Three observable differences between longitudinal sections of specimens D and E [3 marks]

D - Boiled maize seed	E - Fresh viable maize seed
Softer and more swollen after boiling.	Harder, firmer and not swollen.
Embryo/germ may appear cooked, soft or damaged.	Embryo/germ is intact and viable.
May have duller/paler cooked endosperm.	Endosperm appears fresh, firm and natural.
Seed coat may peel more easily.	Seed coat is firmly attached.

2(c)(ii) Two observable similarities between D and E [2 marks]

- Both are maize seeds/grains.
- Both possess seed coat/testa, endosperm and embryo/germ.
- Both contain stored food material.

2(d) Iodine test on crushed D and E [6 marks]

Test	Observation	Conclusion
Iodine solution added to crushed specimen D	Blue-black colour appears	Starch is present in boiled maize seed.
Iodine solution added to crushed specimen E	Blue-black colour appears	Starch is present in fresh viable maize seed.

2(e)(i) Mode of nutrition of specimen F [1 mark]

- Holozoic nutrition; specifically herbivorous feeding on plant seeds/grains.

2(e)(ii) Two economic importance of specimen F [2 marks]

- It causes loss of stored maize and other grains by boring and feeding on them.
- It reduces seed viability, market value and food quality.
- It increases cost of storage and pest control.

2(f)(i) Three storage methods for specimen E [3 marks]

- Dry maize seeds properly before storage.
- Store in clean, dry, airtight containers/sacks/silos.
- Keep in cool, dry and well-ventilated stores.
- Treat seeds with safe recommended insecticide or botanical protectants where appropriate.

- Store above ground on pallets and keep the store clean.

2(f)(ii) Three methods for controlling specimen F [3 marks]

- Dry grains thoroughly to reduce moisture before storage.
- Fumigate or apply recommended storage insecticides by trained persons.
- Use airtight/hermetic storage containers to prevent survival and reproduction.
- Clean stores and remove old infested grains.
- Expose infested grains to sunlight/heat treatment where suitable.

QUESTION 3: SPECIMENS J, K, L, M AND N

J = Cactus plant; K = Waterleaf plant; L = Scapula; M = Humerus; N = Femur.

3(a)(i) One habitat each of specimens J and K [2 marks]

- J - desert/arid or dry habitat; also rocky dry land.
- K - moist/shady habitat, garden soil, damp farmland or wet humus-rich soil.

3(a)(ii) Three adaptations each of J and K to the named habitats [12 marks]

J - Cactus adaptations to dry habitat

- Leaves are reduced to spines, reducing water loss by transpiration and offering protection.
- Thick, fleshy green stem stores water and carries out photosynthesis.
- Thick waxy cuticle reduces evaporation from the stem surface.
- Extensive roots absorb scarce rainwater quickly.
- Sunken/fewer stomata reduce transpiration.

K - Waterleaf adaptations to moist habitat

- Broad, soft leaves expose large surface area for light absorption in shady places.
- Succulent stem and leaves store water.
- Thin cuticle allows easy gaseous exchange/transpiration in moist conditions.
- Shallow/fibrous roots absorb water and mineral salts from moist topsoil.
- Soft herbaceous stem grows rapidly in wet soil.

3(b)(i) Which specimen would the goat preferably feed on? [1 mark]

- Specimen K - Waterleaf plant.

3(b)(ii) Reason [2 marks]

- Waterleaf has soft, succulent, broad leaves and stem and is palatable. Cactus has spines and a tough/waxy stem, making it difficult and unsafe for a goat to feed on.

3(c) Three similarities between specimens J and K [3 marks]

- Both are green plants and can photosynthesize.
- Both possess roots, stems and leaves/modified leaves.
- Both contain chlorophyll.
- Both store water/food in their tissues.
- Both are vascular flowering plants.

3(d)(i) Human parts where L, M and N are found [3 marks]

Specimen	Bone	Part of human body
L	Scapula	Shoulder region/pectoral girdle/upper back
M	Humerus	Upper arm/forelimb
N	Femur	Thigh/hind limb

3(d)(ii) Brief description of specimen L [3 marks]

- Specimen L is the scapula or shoulder blade.
- It is a flat, triangular bone found at the shoulder region.
- It has a spine, acromion/coracoid processes and a glenoid cavity for articulation with the humerus.
- It provides broad surfaces for muscle attachment.

3(d)(iii) How specimen L articulates with specimen M [2 marks]

- The head of the humerus fits into the glenoid cavity of the scapula to form the shoulder joint. This is a ball-and-socket joint that allows movement of the upper arm in many directions.

3(e) Two functions of specimen N in the organism [2 marks]

- Supports the weight of the body and transmits it to the lower limb.
- Provides attachment surfaces for muscles of the thigh and hip.
- Aids locomotion such as walking, running and jumping.
- Protects/supports tissues of the thigh and forms joints with the pelvis and tibia.

EXAMINER'S SUMMARY OF KEY ANSWERS

Item	Correct answer
A	Cockroach
B	Egg case of cockroach
C	Grasshopper
D	Boiled maize seeds - starch present
E	Fresh viable maize seeds - starch present
F	Weevil - pest of stored maize; herbivorous/holozoic nutrition
J	Cactus - dry/arid habitat; spines and succulent stem
K	Waterleaf - moist habitat; broad soft succulent leaves
L	Scapula - shoulder blade
M	Humerus - upper arm bone
N	Femur - thigh bone

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